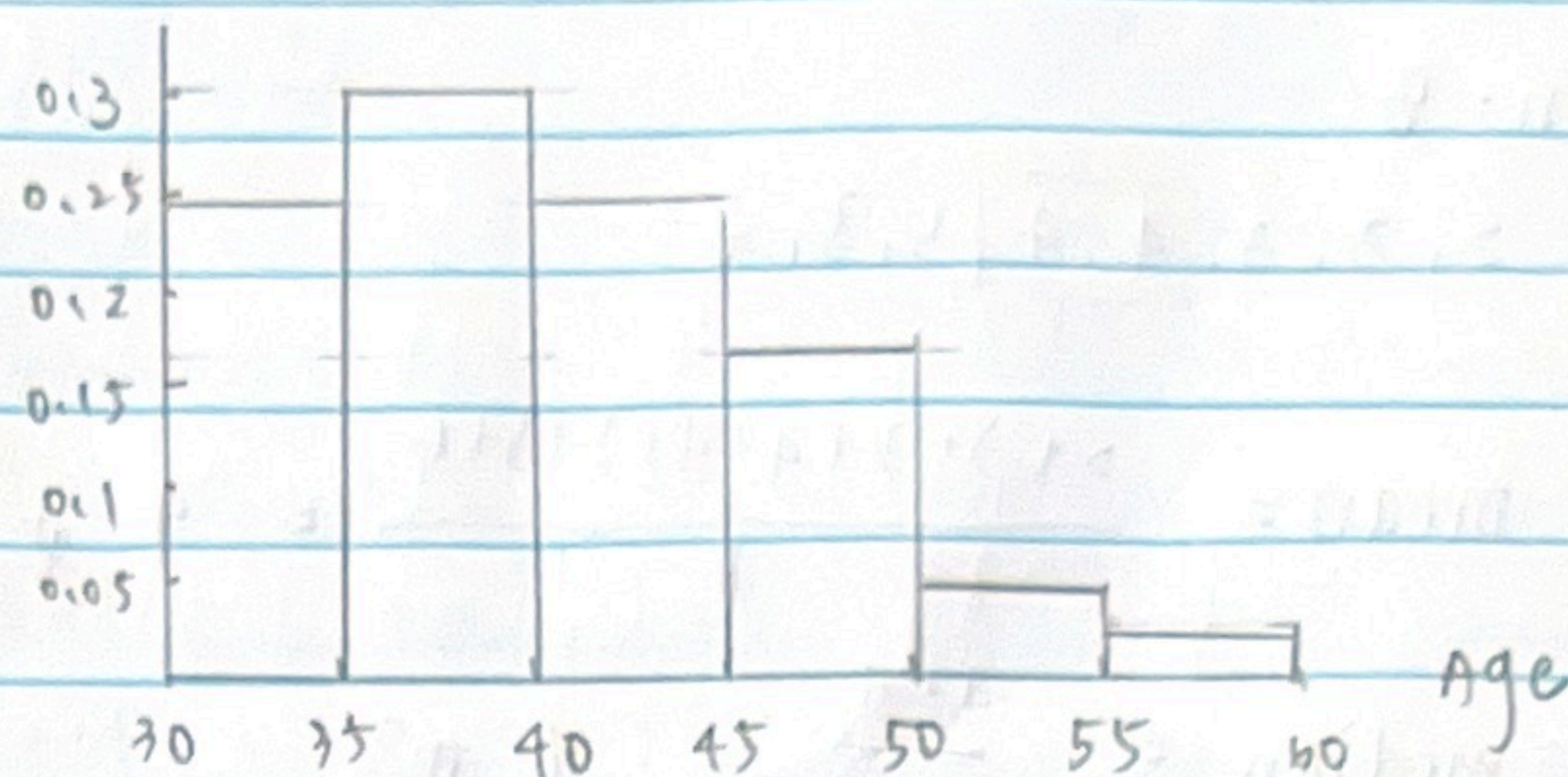


Q 26

a. Histogram

[illegible]

Age	Frequency	Relative Frequency
30-35	12	$12/50 = 0.24$
35-40	15	$15/50 = 0.3$
40-45	12	$12/50 = 0.24$
45-50	8	$8/50 = 0.16$
50-55	2	$2/50 = 0.04$
55-60	1	$1/50 = 0.02$

b. $35 \leq \text{Age} < 45$

$$\Rightarrow \frac{15+12}{50} = 0.54 = 54\%$$

C, Age < 50

$$\Rightarrow \frac{12+15+12+8}{50} = \frac{47}{50} = 0.94 = 94\%$$

CH 2.1

Q2

$n = 8$

2, 3, 3, 4, 4, 5, 5, 6

• mean = $\frac{2+3+3+4+4+5+5+6}{8} = 4$

• median = $\frac{4+4}{2} = 4$

• mode = 3, 4, 5

CH 2.2

Q14

40, 49, 52, 54, 59, 61, 67, 69, 70, 71

Max = 71, Min = 40

a. Range = $71 - 40 = 31$

b. $S = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$ $\bar{x} = \text{mean} = \frac{40+49+52+54+59+61+67+69+70+71}{10} = 59.2$

40	-19.2	368.64
49	-10.2	104.04
52	-7.2	51.84
54	-5.2	27.04
59	-0.2	0.04
61	1.8	3.24
67	7.8	60.84
69	9.8	96.04
70	10.8	116.64
71	11.8	139.24
		<u>967.6</u>

$S = \sqrt{\frac{967.6}{9}} = \sqrt{107.51} \approx 10.37$

c. $\frac{\text{Range}}{S} = \frac{31}{10.37} \approx 3$

CH 2.4

Q2

• mean = $\frac{38.6}{11} = 3.51$

• $z = \frac{x - \bar{x}}{s}$

$z = \frac{1.0 - 3.51}{2.16} = -1.16$

$z = \frac{2.1 - 3.51}{2.16} = -0.65$

$z = \frac{6.5 - 3.51}{2.16} = 1.38$

$z = \frac{2.8 - 3.51}{2.16} = -0.33$

$z = \frac{7.8 - 3.51}{2.16} = 1.99$

$z = \frac{1.7 - 3.51}{2.16} = -0.84$

$z = \frac{2.9 - 3.51}{2.16} = -0.28$

$z = \frac{4.4 - 3.51}{2.16} = 0.41$

$z = \frac{5.1 - 3.51}{2.16} = 0.74$

$z = \frac{2.0 - 3.51}{2.16} = -0.70$

$s = \sqrt{\frac{46.849}{10}} = \sqrt{4.6849} = 2.16$

• z-score

Min z-score $\Rightarrow z = \frac{x - \bar{x}}{s} = \frac{1 - 3.51}{2.16} = \frac{-2.51}{2.16} = -1.16$

Max z-score $\Rightarrow z = \frac{7.8 - 3.51}{2.16} = \frac{4.29}{2.16} = 1.99$

c) 2.4

Q20

1.7, 1.72, 5.9, 8.8, 85.4, 10.1, 11.8
 168, 180, 183, 209, 218, 221, 241
 252, 264, 278, 286, 314, 315, 316
 318, 329, 397, 406, 445, 481, 485

a. • Min = 1.7

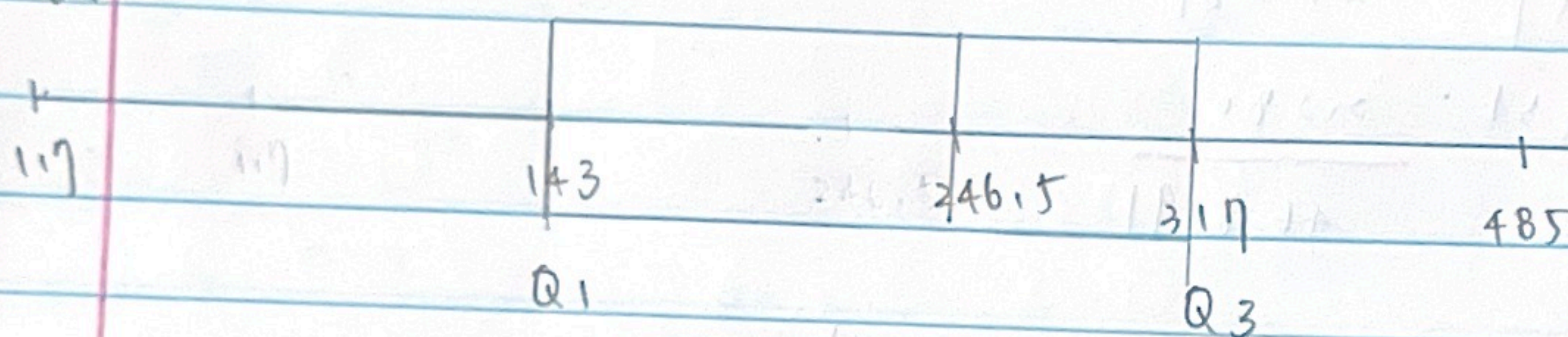
• Median $Q_2 = \frac{241 + 252}{2} = 246.5$

• $Q_1 = \frac{118 + 168}{2} = 143$

• $Q_3 = \frac{316 + 318}{2} = 317$

• Max = 485

b.



c. IQR = $Q_3 - Q_1$

$= 317 - 143 = 174$

lower

$Q_1 - 1.5(174) = 143 - 261 = -118$

Higher

$Q_3 + 1.5(174) = 317 + 261 = 578$

range

$-118 \leq x \leq 578$

⇒ no outliers.

d. Yes, 隨時間累積, 因此成年鯉, 含量應減少

CH 2.4
Q24

233.97 243.92 247.34 252.44 255.4 273.8
286.35 286.41 383.68 408.48 446.3 459.21

$$\bullet \text{ Median} = \frac{273.8 + 286.35}{2} = 280.075$$

$$\bullet Q_1 = \frac{247.34 + 252.44}{2} = 249.89$$

$$\bullet Q_3 = \frac{383.68 + 408.48}{2} = 396.08$$

$$\bullet \text{ Min} = 233.97, \text{ Max} = 459.21$$

$$\text{IQR} = Q_3 - Q_1 \\ = 146.19$$

range:

$$30.605 \leq x \leq 615.305$$

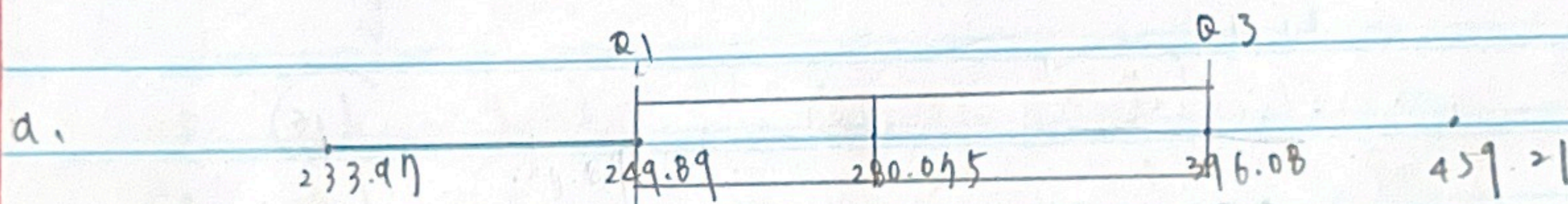
lower

$$Q_1 - 1.5(146.19) = 30.605$$

$\Rightarrow$ NO outlier

higher

$$Q_3 + 1.5(146.19) = 615.305$$

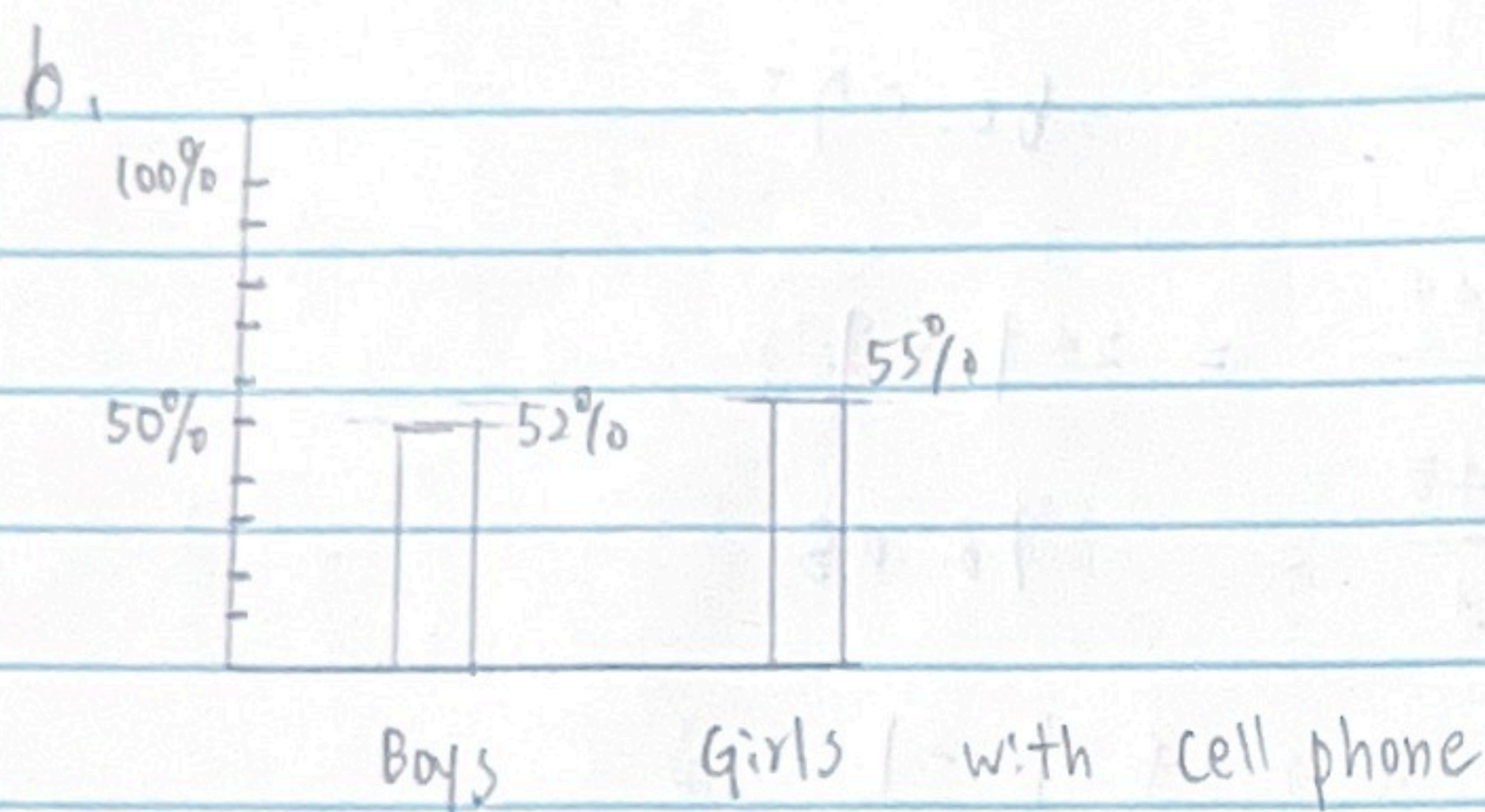


b. right-skewed.

CH 3.1

Q10

	Yes	No
a. Boys	52%	48%
Girls	55%	45%

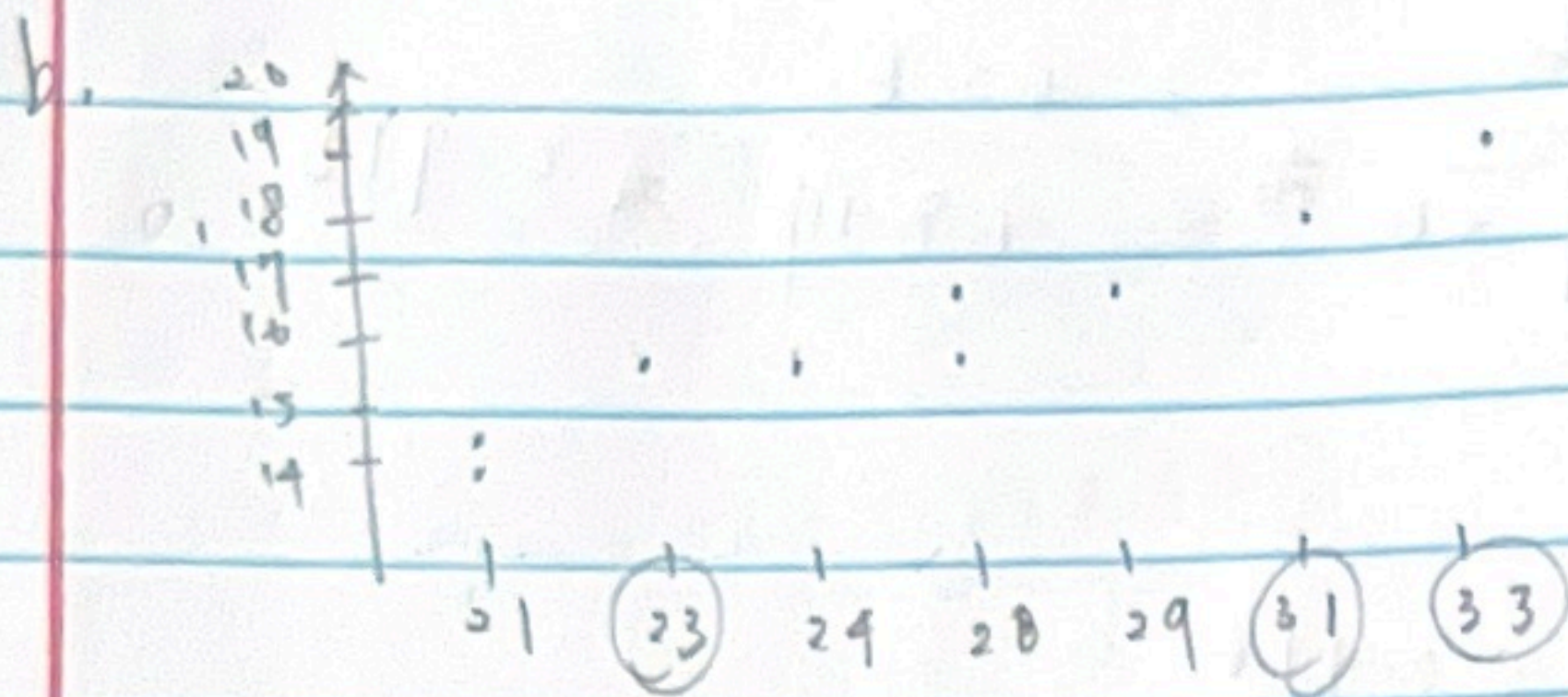


c. Slightly difference between boys and girls.

CH 3.2

Q 25

a. independent variables : Temperature
dependent variables : chips per second



⇒ 越熱叫越長
正相關

c. Least squares line

$$y = a + bx$$

$$\sum x = 21 + 23 + 33 + 29 + 28 + 24 + 21 + 28 + 21 + 28 = 266$$

$$\bar{x} = \frac{266}{10} = 26.6$$

$$\bar{y} = \frac{20 + 16 + 19 + 18 + 18 + 16 + 14 + 17 + 15 + 16}{10} = 16.9$$

$$S_y = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

y			x			multiple
20	3.1	9.61	21	4.4	19.36	13.64
16	-0.9	0.81	23	-3.6	12.96	3.24
19	2.1	4.41	23	6.4	40.96	13.44
18	1.1	1.21	29	2.4	5.76	2.64
18	1.1	1.21	28	1.4	1.96	1.54
16	-0.9	0.81	24	-2.6	6.76	2.34
14	-2.9	8.41	21	-5.6	31.36	16.24
17	0.1	0.01	28	1.4	1.96	6.14
15	-1.9	3.61	21	-5.6	31.36	10.64
16	-0.9	0.81	28	1.4	1.96	-1.26
		30.9			150.4	62.6

$$S_y = \sqrt{\frac{30.9}{9}} \approx 1.833$$

$$S_x = \sqrt{\frac{150.4}{9}} \approx 4.081$$

correlation r

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

$$\Rightarrow \frac{62.6}{\sqrt{(150.4)(30.9)}} = \frac{62.6}{\sqrt{4647.26}} \approx \frac{62.6}{68.17} \approx 0.918$$

$$b = r \frac{s_y}{s_x} \Rightarrow 0.918 \frac{1.853}{4.087} \approx 0.416$$

intercept a:

$$a = \bar{y} - b\bar{x}$$

$$= 16.9 - 0.416(26.6) = 5.83$$

$$\hat{y} = 5.83 + 0.416x$$

d. temperatur. $\rightarrow 17^\circ$

$$\Rightarrow 5.83 + 0.416(27) = 17.062$$

CH 2

Q 2.7

a. Independent variable : $x = \text{Floors}$

dependent variable : $y = \text{Height}$

b.

$$\bar{x} = \frac{1378}{28} = 49.214$$

$$\bar{y} = \frac{5680}{28} = 202.857$$

$$s_y = \sqrt{\frac{51573.428}{27}} = 43.706$$

$$s_x = \sqrt{\frac{2432.714}{27}} = 9.492$$

$$\text{correlation } r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}} = \frac{10714.857}{11201}$$

$$\Rightarrow 0.957$$

c. positive relative.

due to r really close to 1.

d. Least square line

$$\hat{y} = a + bx$$

$$b = \frac{10714.857}{2432.714} = 4.4045$$

$$a = \bar{y} - b\bar{x}$$

$$202.857 - 4.4045(49.214) = -13.91$$

$$\hat{y} = a + bx$$

$$= -13.91 + 4.4045x$$

$$48 \text{ floors} \Rightarrow -13.91 + 4.4045(48) = 197.5 \text{ m}$$